

Role-Based Access Control (RBAC) Specification

This document maps hospital staff roles to specific features and permissions within the Hospital Management System (HMS).

1. The Doctor

The Doctor is the primary clinical user. Their role focuses entirely on patient care, clinical notes, and triggering other workflows (like pharmacy or admission). They do not deal with billing or profits.

- **Appointments:** View their daily schedule and patient queue.
- **Current Status:** Update the patient's status (e.g., *In Consultation*, *Requires Admission*, *Discharged*).
- **Admission & Wards:** View patient history, recommend a patient for admission, and view which cabin their admitted patients are in.
- **Medicine -> Pharmacy:** Write e-prescriptions. The system automatically sends this list to the Pharmacy module.
- **Operation Date + Preparation:** Request an operation, view their surgical schedule, and add pre-op clinical requirements.

2. Receptionist / Front Desk

The first point of contact. Their focus is managing patient flow and bed inventory.

- **Appointment:** Create, reschedule, and cancel patient visits.
- **Admission:** Process the paperwork to admit a patient.
- **Ward, Cabin, Empty Cabin (list):** View the real-time list of empty beds to assign to newly admitted patients.
- **Current Status:** Update status to *Arrived* or *Waiting*.

3. Pharmacist

Focuses purely on dispensing medication and managing inventory.

- **Medicine -> Pharmacy:** Receive the doctor's digital prescription and dispense the items.
- **Bill (Pharmacy):** Generate the sub-bill for medicines.
- **Discount:** Apply basic, standard discounts to medicines (usually up to a small, pre-set limit like 5%).

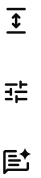
4. Billing / Cashier

Handles all financial transactions for the patient before discharge.

- **Bill:** Consolidate all sub-bills (consultation, cabin rent, pharmacy, operation fees) into a master bill.
- **Discount:** Apply approved discounts to the final bill. (Requires Admin override for large discounts).
- **Current Status:** Mark patient as *Cleared for Discharge* once the bill is paid.

5. OT (Operation Theater) Manager / Head Nurse

Manages the logistics of the surgical department.



- **Operation Date + Preparation:** Book the physical OT rooms, schedule the anesthesiologist, and manage the pre-surgery preparation checklist (fasting, blood arranged, etc.).
- **Current Status:** Update status to *In Surgery* and *Recovery*.

6. Admin / Management

The system owner. They have a bird's-eye view of the hospital and control security and finances.

- **Dashboard:** View daily metrics (total admissions, total revenue, pending surgeries).
- **Admin Report (Range) + Profit:** Generate financial reports across custom date ranges to calculate exact profit margins.
- **Role Management:** Create new user accounts, reset passwords, and assign system roles (e.g., promoting a user to OT Manager).
- **Discount (Approval):** Override standard limits to approve high-value or VIP discounts.